

Extra Hidden States under Emission Misspecification in Hidden Markov Models

Aurélien Nicosia

Département de mathématiques et de statistique, Université Laval

Québec, Québec, Canada

`aurelien.nicosia@mat.ulaval.ca`

ORCID: [0000-0002-0974-9988](https://orcid.org/0000-0002-0974-9988)

Abstract

Additional hidden Markov states can compensate for restrictive emission distributions without representing additional scientific regimes. Building on the established representation of mixture components as refined states, we prove a uniform lower bound on the observed-law Kullback–Leibler divergence that grows linearly with sequence length. The result assumes a compact, identifiable emission family, positive aggregate transitions and distinct mixture components. An exact refinement removes this approximation gap. Simulations examine state splitting and its sensitivity to numerical optimization. In tumor allele-fraction data, five Gaussian components describe three regional allelic profiles using fourteen fewer parameters than a five-state Gaussian model, with comparable held-out prediction. Technical replication supports the broad profiles, while short segments remain sensitive to preprocessing. A symmetry-aware comparator has lower BIC but mixed predictive advantages. A parametric bootstrap does not reject the refinement constraints at the 5% level (p-value 0.09); independent sequencing limits the interpretation of aggregate states.

Keywords: hidden Markov model; emission misspecification; finite mixture; relative-entropy rate; state splitting; allelic imbalance.

MSC 2020: Primary 62M05; Secondary 62M10.

1 Introduction

Hidden Markov models (HMMs) separate serial dependence from the conditional distribution of observations through a latent state process. This decomposition supports applications across time-series analysis, signal processing, and ecology (Rabiner, 1989; Cappé et al., 2005; Zucchini et al., 2016). Fitted states are often labelled as behaviours, regimes, phases, or classes and then become the main scientific object reported from an analysis (McClintock et al., 2020).

That interpretation is useful but not automatic. A fitted state is also part of a statistical approximation. If the state-dependent distribution is too restrictive, likelihood can improve by

creating states whose emissions jointly approximate skewness, heavy tails, overdispersion, or within-regime heterogeneity. An additional state can therefore be statistically useful without representing an additional scientific regime. Under misspecification, likelihood targets a best approximation in the fitted family rather than the data-generating parameter (White, 1982; Douc and Moulines, 2012).

The practical phenomenon is established in the HMM literature. Pohle et al. (2017) show that information criteria can favour state counts that are difficult to interpret scientifically. Additional states may absorb unmodelled structure in state-dependent distributions, while flexible parametric, mixture, and nonparametric observation models offer alternatives (Dannemann, 2012; Dannemann et al., 2014; Langrock et al., 2015, 2018). The distinction matters because methods for estimating statistical HMM order (Gassiat and Keribin, 2000; Gassiat and Rousseau, 2014; Lehericy, 2019; Zou et al., 2024; Lin and Huang, 2025) do not imply that every selected state is a distinct scientific regime.

Mixture-emission estimation and component aggregation already have a substantial methodological basis. Volant et al. (2014) give the component-state representation and an EM-based merging approach. Holzmänn and Schwaiger (2015) characterize minimal representations and test whether states can be reduced to mixture components without changing the observed law. We use this established refinement construction as the starting point, not as a new model or estimation algorithm.

Our result concerns uniform separation from a restrictive emission class. Under compactness, identifiable mixture order and positive aggregate transitions, every one-step true predictive density remains a positive L^1 distance from every mixture with only the aggregate number of simple components. The chain rule turns this bound into a Kullback–Leibler divergence that is at least proportional to sequence length against every HMM in the smaller class, without aligning hidden paths or restricting candidate transitions. Marginal mixture identifiability already establishes non-representability. The additional predictive bound gives uniform separation at every sequence length and a positive relative-entropy-rate gap. The result complements HMM likelihood and identifiability theory (Leroux, 1992; Bickel et al., 1998; Allman et al., 2009; Gassiat et al., 2016; Alexandrovich et al., 2016).

The empirical comparison concerns the allocation of distributional complexity to emissions or to latent dynamics. We compare simple-emission and mixture-emission HMMs, including pairs with the same total number of Gaussian components. The Bayesian information criterion (BIC), held-out sequential predictions, and decoded paths assess different aspects of this comparison. A simulation isolates the mechanism. The application examines unphased tumor allele fractions, for which the measurement process gives a direct reason for paired emission components. It asks whether low- and high-fraction bands require distinct latent transition laws. Held-out chromosomes, a technical replicate, symmetry-aware models and an independent assay assess the resulting regional interpretation. The Supporting Information also examines elk movement, including its sensitivity to the emission family and the distinction between aggregate states and density modes.

Section 2 gives the formal mechanism; Sections 3 and 4 describe estimation and model comparison; and Sections 5 and 6 illustrate the comparison in simulation and data.

2 Why restrictive emissions create extra states

The practical message of this section is simple. If a scientifically meaningful regime has a heterogeneous emission distribution, several simple-emission states can improve fit by representing its components rather than by revealing new regimes. The formal results below give conditions under which an aggregate-state model with richer emissions and a refined model with additional simple-emission states generate the same observed process.

Let $\mathbf{Y}$ be the observation set, let $\mathcal{B}$ be its sigma-field, let λ be a dominating measure on $(\mathbf{Y}, \mathcal{B})$, and let Θ be the emission-parameter space. All densities below are with respect to λ . For densities p and q , define the Kullback–Leibler divergence and the $L^1(\lambda)$ distance by

$$D(p\|q) = \int_{\mathbf{Y}} p(y) \log \left\{ \frac{p(y)}{q(y)} \right\} \lambda(dy), \quad \|p - q\|_1 = \int_{\mathbf{Y}} |p(y) - q(y)| \lambda(dy).$$

Here, $0 \log(0/q) = 0$, and $D(p\|q) = +\infty$ if $q = 0$ on a set having positive probability under p . The same notation $D(P\|Q)$ is used for probability laws through their densities. Let $\mathcal{F} = \{f_\eta : \eta \in \Theta\}$ be the simple fitted emission family. Define the class of mixtures with at most K atoms, for an integer $K \geq 1$, by

$$\mathcal{M}_K = \left\{ \sum_{j=1}^K \omega_j f_{\eta_j} : \omega_j \geq 0, \sum_{j=1}^K \omega_j = 1, \eta_j \in \Theta \right\}.$$

Thus, the ω_j are mixing proportions. Zero weights are permitted, so $\mathcal{M}_K \subseteq \mathcal{M}_{K+1}$. Let $\mathbf{H}_K(\mathcal{F})$ denote the observed process laws generated by K -state HMMs with emissions in $\mathcal{F}$, arbitrary transition matrices, and arbitrary initial distributions. For process laws P and Q , let $P^{1:T}$ and $Q^{1:T}$ denote the respective joint laws of $(Y_1, \dots, Y_T)$ and write

$$\underline{D}^{\text{rate}}(P\|Q) = \liminf_{T \rightarrow \infty} T^{-1} D(P^{1:T}\|Q^{1:T})$$

for their lower relative-entropy rate.

Fix an integer $K_0 \geq 1$, interpreted as the number of aggregate scientific regimes. A superscript 0 denotes a data-generating parameter. The following assumptions isolate a transparent finite-mixture setting.

Assumption 2.1 (Emission regularity). The parameter space Θ is compact, the map $\eta \mapsto f_\eta$ is continuous into $L^1(\lambda)$, and every f_η is positive λ -almost everywhere.

Assumption 2.2 (Identifiable mixture order). Finite mixtures from $\mathcal{F}$ have identifiable minimal order. Two equal finite mixtures with positive weights and pairwise distinct parameters have the same number of components and agree up to permutation.

Classical sufficient conditions are given by [Teicher \(1963\)](#) and [Yakowitz and Spragins \(1968\)](#); see also [McLachlan and Peel \(2000\)](#). The assumptions include univariate Gaussian emissions on compact mean and variance sets with variances bounded away from zero.

Lemma 2.3 (Separation from lower-order mixtures). *Under Assumptions 2.1 and 2.2, let $R > K_0$ be an integer and let $\eta_1^0, \dots, \eta_R^0 \in \Theta$ be pairwise distinct. For $c > 0$ and $Rc \leq 1$, set*

$$\mathcal{A}_c = \left\{ \sum_{r=1}^R \omega_r f_{\eta_r^0} : \omega_r \geq c, \sum_{r=1}^R \omega_r = 1 \right\}.$$

Then $d_c := \inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} \|p - q\|_1 > 0$, and

$$\inf_{p \in \mathcal{A}_c} \inf_{q \in \mathcal{M}_{K_0}} D(p \| q) \geq d_c^2 / 2 > 0.$$

The lemma says that a mixture containing R distinct atoms, each with a non-negligible weight, cannot be approximated arbitrarily closely by only $K_0 < R$ atoms.

To see the approximation incentive directly, let $Z_t \in \{1, \dots, K_0\}$ be a stationary aggregate Markov chain with transition matrix $\Gamma^0 = (\gamma_{k\ell}^0)$ and stationary probabilities $\pi_k = \mathbb{P}(Z_t = k) > 0$. Here, $\gamma_{k\ell}^0 = \mathbb{P}(Z_t = \ell \mid Z_{t-1} = k)$. Suppose the emission in state k is g_k , and, for an integer $m \geq 1$, define $\delta_k(m) = \inf_{q \in \mathcal{M}_m} D(g_k \| q)$. Thus, $\delta_k(m)$ is the infimum of the emission-level Kullback–Leibler loss with at most m atoms from $\mathcal{F}$. For specified aggregate-state emissions $q_1, \dots, q_{K_0}$, define the aligned risk by

$$\mathcal{R}(q_1, \dots, q_{K_0}) = \sum_{k=1}^{K_0} \pi_k D(g_k \| q_k).$$

This comparison fixes the law of the aggregate hidden trajectory. In the refined representation, the within-state component is drawn independently at each visit, conditional on the aggregate states; the emission q_k can therefore be any member of $\mathcal{M}_{m_k}$.

Proposition 2.4 (Aligned improvement by state splitting). *If aggregate state k is represented by one emission from $\mathcal{F}$, the infimum of the aligned Kullback–Leibler risk is $\sum_{k=1}^{K_0} \pi_k \delta_k(1)$. If it is represented by $m_k \geq 1$ substates, where m_k is the number allocated to aggregate state k , the infimum of the aligned risk is $\sum_{k=1}^{K_0} \pi_k \delta_k(m_k)$. If $\sum_k \pi_k \delta_k(1) < \infty$, the second risk is strictly smaller whenever $\delta_{k^*}(m_{k^*}) < \delta_{k^*}(1)$ for at least one state.*

In plain terms, splitting gives a heterogeneous state several emission atoms instead of one. The statement is only an aligned comparison because it holds the aggregate trajectory law fixed; attainment of the infima is not required. The next construction removes that restriction.

We recall the component-state construction of [Volant et al. \(2014\)](#); see also [Holzmann and Schwaiger \(2015\)](#). Suppose each aggregate emission is a finite mixture with m_k components and $\eta_{k,a}^0 \in \Theta$,

$$g_k(y) = \sum_{a=1}^{m_k} \omega_{k,a}^0 f_{\eta_{k,a}^0}(y), \quad \omega_{k,a}^0 > 0, \quad \sum_a \omega_{k,a}^0 = 1. \quad (1)$$

The $\omega_{k,a}^0$ are the within-state mixing proportions.

Proposition 2.5 (Exact HMM refinement). *Let the refined states be (k, a) , where $1 \leq k \leq K_0$ and $1 \leq a \leq m_k$. Give state (k, a) emission $f_{\eta_{k,a}^0}$ and transition probabilities*

$$\Gamma_{(k,a),(\ell,b)}^{\text{ref}} = \gamma_{k\ell}^0 \omega_{\ell,b}^0. \quad (2)$$

With initial probabilities $\pi_k \omega_{k,a}^0$, the refined HMM has exactly the same observed process law as the aggregate HMM with transitions Γ^0 and emissions in (1).

Equation (2) retains the probability of moving from aggregate state k to ℓ , then draws a component of the destination emission. The refined states therefore encode within-regime mixture components, not new aggregate dynamics.

Assume now that $\varepsilon := \min_{1 \leq k, \ell \leq K_0} \gamma_{k\ell}^0 > 0$, let $R = \sum_{k=1}^{K_0} m_k$, let $\omega_{\min} = \min_{1 \leq k \leq K_0, 1 \leq a \leq m_k} \omega_{k,a}^0$, and set $c = \varepsilon \omega_{\min}$. Suppose $R > K_0$ and all $\eta_{k,a}^0 \in \Theta$ are pairwise distinct.

Theorem 2.6 (Observed-law separation). *Under Assumptions 2.1 and 2.2, the stationary aggregate HMM with emissions (1) has observed law P_0^Y . For every integer $T \geq 1$ and every $Q \in \mathcal{H}_{K_0}(\mathcal{F})$,*

$$D\{(P_0^Y)^{1:T} \| Q^{1:T}\} \geq T d_c^2/2. \quad (3)$$

Consequently,

$$\inf_{Q \in \mathcal{H}_R(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0, \quad \inf_{Q \in \mathcal{H}_{K_0}(\mathcal{F})} \underline{D}^{\text{rate}}(P_0^Y \| Q) \geq d_c^2/2 > 0.$$

The constant is uniform over the candidate HMMs, but depends on the fixed true atoms, their minimum weight, the transition bound and the compact fitted parameter set. It need not stay bounded away from zero when distinct atoms approach one another or these lower bounds vanish.

The first equality follows because the refined R -state model reproduces the truth exactly. For the second, every one-step prediction from a K_0 -state simple-emission HMM belongs to $\mathcal{M}_{K_0}$, while every true prediction contains all R atoms with weights at least c . The mixture separation lemma and the chain rule for relative entropy give (3) for every T , hence the positive rate gap. This argument concerns observed predictive distributions and does not align hidden paths.

Corollary 2.7 (Enriched-emission representation). *If the fitted emission family is enlarged from $\mathcal{F}$ to $\mathcal{M}_M$, with an integer $M \geq \max_{1 \leq k \leq K_0} m_k$, then*

$$\inf_{Q \in \mathcal{H}_{K_0}(\mathcal{M}_M)} \underline{D}^{\text{rate}}(P_0^Y \| Q) = 0.$$

Thus, a refined model with more simple-emission states and an aggregate model with richer emissions can describe the same observations. Equality of the two representations does not determine which latent description is scientifically meaningful.

A likelihood-selection consequence requires an additional stochastic limit, not just a bound on expected divergence. For example, suppose the normalized log-likelihood ratios $T^{-1} \log\{p_0(Y_{1:T})/q(Y_{1:T})\}$ converge almost surely, uniformly over $Q \in \mathbf{H}_{K_0}(\mathcal{F})$, to limits bounded below by $d_c^2/2$. Then the maximized log-likelihood advantage of the R -state class is at least $T(d_c^2/2 - o(1))$, since this class contains the truth. Any fixed parameter-count BIC penalty difference is only of order $\log T$. Under that additional assumption, BIC eventually prefers R to K_0 simple-emission states. This conditional comparison neither proves the uniform convergence assumption nor selects R against every other order.

For reference, Sections S1.1–S1.5 of the Supporting Information give proofs of Lemma 2.3, Proposition 2.4, Proposition 2.5, Theorem 2.6, and Corollary 2.7, respectively.

3 Fitting the enriched-emission HMM

Let $X_t \in \{1, \dots, K\}$ denote the fitted hidden state at time t . For state k , let $M_k \geq 1$ be the number of fitted Gaussian components and let $\phi(y; \mu, \sigma^2)$ denote the Gaussian density at y with mean μ and variance $\sigma^2 > 0$. The fitted state-dependent density is

$$b_k(y) = \sum_{m=1}^{M_k} \omega_{km} \phi(y; \mu_{km}, \sigma_{km}^2), \quad \omega_{km} \geq 0, \quad \sum_m \omega_{km} = 1.$$

The quantities ω_{km} are mixing proportions within state k ; they do not change the state transition probabilities. The hidden chain has initial probabilities $\alpha_k = \mathbb{P}(X_1 = k)$ and transition probabilities $a_{k\ell} = \mathbb{P}(X_t = \ell \mid X_{t-1} = k)$, collected in $A = (a_{k\ell})$.

The following standard mixture-emission expectation-maximization (EM) updates are included to specify the fitted model (Volant et al., 2014). The algorithm alternates between state and component allocation and closed-form parameter updates. Write $Y_{1:T} = (Y_1, \dots, Y_T)$ and let $C_t \in \{1, \dots, M_{X_t}\}$ denote the Gaussian-component label at time t . A scaled forward–backward recursion gives

$$\begin{aligned} \zeta_t(k) &= \mathbb{P}(X_t = k \mid Y_{1:T}), \\ \xi_t(k, \ell) &= \mathbb{P}(X_{t-1} = k, X_t = \ell \mid Y_{1:T}). \end{aligned}$$

Here, $\zeta_t(k)$ is defined for $t = 1, \dots, T$, whereas $\xi_t(k, \ell)$ is defined for $t = 2, \dots, T$. The joint posterior responsibility for state k and component m is

$$r_t(k, m) = \mathbb{P}(X_t = k, C_t = m \mid Y_{1:T}) = \zeta_t(k) \frac{\omega_{km} \phi(Y_t; \mu_{km}, \sigma_{km}^2)}{b_k(Y_t)}.$$

The M-step updates are

$$\begin{aligned}\alpha_k^{\text{new}} &= \zeta_1(k), & a_{k\ell}^{\text{new}} &= \frac{\sum_{t=2}^T \xi_t(k, \ell)}{\sum_{t=2}^T \sum_j \xi_t(k, j)}, \\ \omega_{km}^{\text{new}} &= \frac{\sum_t r_t(k, m)}{\sum_t \zeta_t(k)}, & \mu_{km}^{\text{new}} &= \frac{\sum_t r_t(k, m) Y_t}{\sum_t r_t(k, m)}, \\ (\sigma_{km}^2)^{\text{new}} &= \frac{\sum_t r_t(k, m) (Y_t - \mu_{km}^{\text{new}})^2}{\sum_t r_t(k, m)}.\end{aligned}$$

In these updates, sums over t without displayed limits run from 1 to T , and $\sum_j$ runs over the K fitted states. For independent sequences with common parameters, the forward-backward recursion resets at each sequence start. The initial-probability update is the average of the state responsibilities at those starts; transition sums exclude boundaries, while emission sums pool all observations. Numerical pseudocounts and variance constraints are specified in SI S2–S3. In computation, a small lower bound on σ_{km} prevents likelihood degeneracy. Multiple starts are needed because mixture-HMM likelihoods are multimodal. For a sequence of length T , BIC is $-2\ell(\hat{\theta}) + p \log T$, where $\hat{\theta}$ is the maximum likelihood estimate, $\ell(\hat{\theta})$ is the maximized log-likelihood, and the number p of free parameters is

$$(K-1) + K(K-1) + \sum_{k=1}^K \{(M_k-1) + 2M_k\}.$$

The three terms count initial probabilities, transition probabilities, and state-specific mixture parameters, respectively.

4 Comparing emission and transition complexity

Suppose scientific knowledge suggests K_0 aggregate regimes, while a simple-emission analysis favours additional states. A K_0 -state HMM with $R = \sum_k M_k$ Gaussian components is an R -state Gaussian HMM whose transition probabilities satisfy the refinement constraints in Proposition 2.5, with initial probabilities $\alpha_{(k,m)}^{\text{ref}} = \alpha_k \omega_{km}$. Comparing it with an unrestricted R -state model holds the emission component count fixed and changes the latent dynamics. For example, two aggregate states with component counts $(2, 1)$ have 10 free parameters, compared with 14 for three Gaussian states; counts $(2, 2)$ give 13, compared with 23 for four Gaussian states. A comparison of two states with four components against three Gaussian states changes both the component count and the transition structure.

The factorization requires both equal transition rows within each source group and destination-component proportions that do not depend on the source state. Equal rows or a low matrix rank alone do not establish the full constraint (Holzmann and Schwaiger, 2015). Here the groupings are specified from the emission mechanism. The state-reduction likelihood-ratio test of Holzmann and Schwaiger (2015) addresses these constraints. Under their regularity conditions and stationary initial

law, reducing k states to r groups gives $k^2 - 2k - r^2 + 2r$ reference degrees of freedom, hence 12 for $k = 5$ and $r = 3$. Our genomic fits instead estimate a common initial law from two independent sequences. The information about this initial law does not grow as it does for transition parameters when only sequence lengths grow; adding two initial-probability parameters therefore does not by itself justify a chi-squared law with 14 degrees of freedom. We use a parametric bootstrap under the fitted constrained model, retaining the observed sequence lengths, the same screening and refinement settings for observed and simulated data (SI S3.2), and starts at the exact refinement. This calibration is conditional on the fitted model.

BIC summarizes penalized likelihood within the specified model classes. It is not a test of the number of biological regimes. In particular, mixture singularities and transition estimates near the boundary limit a regular asymptotic interpretation of its penalty. An unrestricted refined fit must have likelihood at least as high as the corresponding constrained fit at a common variance bound; initialization at the exact refinement provides a numerical check of that nesting.

Prediction is assessed separately. Parameters estimated from a training prefix $Y_{1:n}$ are held fixed while the filter is updated using subsequent observations. For a validation block $n+1, \dots, n'$, the log score is $\sum_{t=n+1}^{n'} \log p_{\hat{\theta}_n}(Y_t | Y_{1:t-1})$. The probability integral transforms $U_t = F_{\hat{\theta}_n}(Y_t | Y_{1:t-1})$ assess the predictive distribution, rather than only its average log density. Exact continuous conditional distributions give uniform PIT values; fitted models need not. Empirical PIT discrepancies and central interval coverage are descriptive checks here, without an independence-based significance test.

Decoded sequences and state-dependent distributions connect these comparisons to the observations. Cross-tabulations indicate when two Gaussian states map largely to one mixture-emission state, but do not establish that the latter is scientifically correct. States index Markov dynamics; mixture components describe within-state distributions; modes are local maxima of a density. These three counts need not coincide. Interpretation therefore also depends on temporal stability, sensitivity to fitting constraints, and measurements relevant to the scientific question.

5 Simulation study

We first isolate the mechanism in a controlled setting. The data-generating process has two aggregate states, transition matrix

$$\Gamma^0 = \begin{pmatrix} 0.93 & 0.07 \\ 0.05 & 0.95 \end{pmatrix}.$$

The first-state emission was a two-component Gaussian mixture with weights 0.70 and 0.30, means -1.20 and 1.10 , and standard deviations 0.55 and 0.45 . The second-state emission was Gaussian with mean 3.80 and standard deviation 0.65 . The process therefore has two aggregate regimes and three distinct Gaussian atoms.

For a reference series of length 1500, we fitted Gaussian HMMs with $K = 2$ and $K = 3$, and an enriched $K = 2$ HMM with two Gaussian components per state. Each model was fitted with the

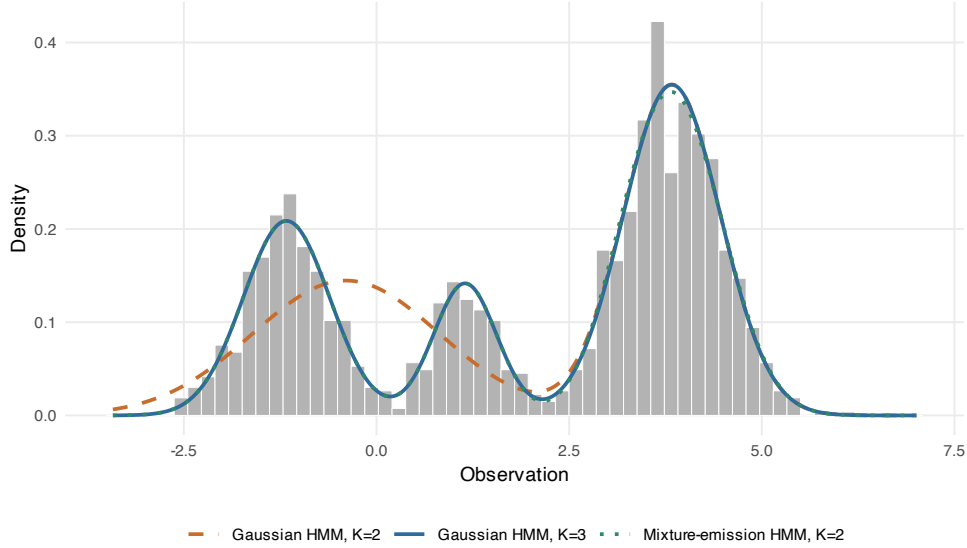


Figure 1: Observed histogram and fitted marginal densities for the reference simulation. The two-state Gaussian model smooths over the heterogeneous first regime. The three-state Gaussian and enriched two-state models recover its two components.

protocol used for the replications below. The two-state Gaussian model merged the heterogeneous first regime and had BIC 4487.4. The three-state Gaussian model separated its two components and had BIC 4197.3. The enriched two-state model recovered the same within-regime structure while retaining two aggregate states and had BIC 4191.6. [Figure 1](#) shows the directly observable emission fit.

Across 200 fixed-seed replications at each sample size, BIC preferred the three-state to the two-state Gaussian model in 99.5%, 100% and 100% of samples for $T = 500, 1500, 5000$, respectively. Comparing G3 with the enriched two-state model, BIC preferred the latter in 96.0%, 99.0% and 99.5%. Both classes contain the data-generating law. Their fitted log likelihoods are nearly equal on average, and the mean M22-minus-G3 BIC differences $-6.5, -7.7, -8.6$ are close to $-\log T$. The preference mainly reflects M22’s smaller parameter count (13 versus 14), rather than a substantial improvement in fit. Each model retains the best fit from the original random starts and additional data-derived starts. In particular, the exact refinement of a fitted M21 model supplies a G3 start, without using generating parameters. Relative likelihood convergence is checked at 10^{-9} . The more thorough optimization reverses 39 of the 40 original G2–G3 BIC preferences for two states; those original failures primarily reflected local maxima rather than state-order selection. SI S2 documents the protocol and retained fits. The enriched candidate has four components although the data-generating distribution has three; these frequencies therefore do not isolate the equal-component comparison of Section 4. That comparison is implemented directly in the genomic analysis. Full computational details and additional diagnostics are reported in the Supporting Information.

These frequencies are descriptive. Finite mixtures and overfitted HMMs contain singular parameter strata, so regular-model interpretations of BIC require caution ([Keribin, 2000](#); [Rousseau](#)

and Mengersen, 2011; Gassiat and Rousseau, 2014; Watanabe, 2013; Drton and Plummer, 2017).

6 Chromosomal allelic imbalance

6.1 Scientific question and measurements

We analyze HCC1143, a breast-cancer cell line (cells maintained in laboratory culture), and its matched normal cell line HCC1143BL, using the SNP-array data accompanying [Chiang et al. \(2009\)](#). The aim is to describe chromosomal regions with similar relative contributions of DNA inherited from the two parents. A region with unequal contributions may produce two bands of observed fractions ([Olshen et al., 2011](#)).

A single-nucleotide polymorphism (SNP) is a DNA position at which alternative single-base variants occur. The two variants assayed at each SNP are labelled A and B. A marker is heterozygous when both variants are present in the matched normal. The array supplies signal intensities for the two alleles; the B-allele fraction is the B signal divided by the combined A and B signals ([Bengtsson et al., 2010](#)). Approximately equal contributions give a fraction near one half; unequal contributions produce a low or high fraction.

Haplotype phasing identifies which variants lie on the same inherited chromosome copy. These array measurements are unphased: the allele designated B need not belong to the same parental chromosome at successive markers ([Chen et al., 2017](#)). For illustration, fractions near 0.2 and 0.8 at different markers can describe the same regional imbalance. Switching between these bands need not mark a regional change. This is the measurement-based reason for a mixture emission.

The statistical question is whether the low- and high-fraction bands require separate transition laws or can be represented within regional allelic profiles. Here a profile describes balance or a degree of imbalance, not a cell population. DNA copy number is the total number of copies of a region; allele fractions alone do not determine it. Equal contributions can occur with different total copy numbers ([Olshen et al., 2011](#)).

The data are deposited in Gene Expression Omnibus (GEO), series [GSE13372](#), which contains repeated Affymetrix GenomeWideSNP 6.0 array measurements. We selected the first two accession-ordered tumor-normal replicate pairs before inspecting their processed profiles: GSM337641/GSM337662 and GSM337642/GSM337663. CEL files store the original probe intensities. AS-CRMAv2 converts these to allele-specific signals; TumorBoost then uses the matched normal to reduce marker-specific distortion of tumor fractions ([Bengtsson et al., 2009, 2010](#)). We retain candidate heterozygous markers with matched-normal fractions in $[0.3, 0.7]$, finite measurements and unique positive genomic positions. The selection rule does not use the tumor fraction: the normal sample identifies markers informative about allelic imbalance without selecting them on the tumor response. Each HMM observation is one corrected tumor fraction at a retained marker. Corrected values are not clipped at zero or one.

Chromosome 1 of the first pair supplies 20,697 markers for estimation. Chromosomes 12–17 supply 52,294 validation markers. After the fitted parameters and specification were frozen,

chromosomes 18–22 supplied a further 25,413 confirmation markers. Chromosomes 2–11 were not used. The second pair provides a technical replication on chromosome 1, with 20,937 markers, including 19,727 shared with the estimation series. Technical replication means measuring the same cell lines again, rather than sampling additional patients. All coordinates use the hg18 human reference-genome assembly. Each chromosome boundary and gap exceeding one megabase (Mb, one million base pairs) resets the chain; chromosome 1 contains two sequences. The index orders markers by position, rather than representing equal elapsed times or equal distances.

6.2 Models and estimation

We fit ordinary Gaussian HMMs with two, three and five states (G2, G3 and G5). M21 has two aggregate states with component counts (2, 1); M221 has three with counts (2, 2, 1). The latter can represent two degrees of imbalance and an approximately balanced profile. These are specified candidate representations, not a claim that three is the true or globally optimal number of genomic states. The M21–G3 and M221–G5 comparisons hold the total number of Gaussian components fixed. G5 relaxes the transition restrictions of the five-state refinement of M221, adding 14 parameters without adding emission components.

Each unconstrained family uses 40 random starts, supplemented by persistent variance-regime starts for G2 and paired-band starts for the mixture models. The eight highest likelihoods are refined to a relative likelihood tolerance of 10^{-8} , with a maximum of 2,500 iterations. Exact refinements and nested duplicated-component starts check the likelihood inclusions. All selected fits converge, and none reaches the standard-deviation floor of 0.01. Competing solutions, transition estimates and numerical checks are reported in Supporting Information, Section S3.

A symmetry-aware comparison uses state densities

$$f_k(y) = \frac{1}{2}\phi(y; \frac{1}{2} - d_k, s_k^2) + \frac{1}{2}\phi(y; \frac{1}{2} + d_k, s_k^2), \quad d_k \geq 0.$$

The two- and three-state versions, F2 and F3, are equivalent to folded-normal models for $|y - 1/2|$. They have four and six tied Gaussian components and 7 and 14 free parameters, respectively. F3 is a separate scientific comparator, not the equal-component or a nested counterpart of M221. Their predictive densities are evaluated on the original fraction scale for comparison with the other models. This incorporates known allelic symmetry, instead of requiring an unconstrained mixture to estimate it. We use 30 starts and refine the best six. The BIC values describe these fitted candidates; we do not invoke a regular likelihood-ratio reference law or a general consistency theorem for mixture order selection.

6.3 Components and regional profiles

M221 estimates paired component centres at (0.185, 0.805) for strong imbalance and (0.392, 0.601) for moderate imbalance, and a single centre at 0.498 for approximate balance. The within-pair weights are close to one half. Posterior occupations are 38.3%, 49.2% and 12.5%, and self-transition

probabilities are 0.988, 0.998 and 0.963, respectively. The names describe the fitted fraction patterns, without translating their centres into integer copy numbers. Figure 2 shows their spatial organization and separates the component densities from the aggregate profiles.

Table 1: Chromosome-1 estimates and fixed-parameter held-out log scores. Lower BIC and higher log scores are preferred within their respective columns. The Replicate column uses the second array pair on chromosome 1, a technical repeat of the same cell line; it is not biological replication. The chromosome blocks concern that cell line. F2/F3 impose reflection symmetry.

Model	p	BIC	Replicate	Chr. 12–17	Chr. 18–22
G2	7	-14751.5	7049.59	-46.97	15572.46
M21	10	-20773.1	9816.15	14423.18	17015.30
G3	14	-20738.2	9818.62	14410.44	17013.16
M221	20	-24817.8	11906.57	16386.29	21371.08
G5	34	-24697.0	11907.57	16350.04	21365.91
F2	7	-21890.9	10309.49	14620.20	16146.25
F3	14	-24835.2	11887.64	16379.29	21386.96

G5 estimates almost the same five component distributions. Its maximized log likelihood exceeds M221’s by 9.13, whereas its BIC is 120.87 higher (Table 1). Its low- and high-fraction states within each imbalance pair have nearly equal transition rows: their total-variation distances are 0.0104 and 0.0065. The five transition-matrix singular values are approximately 1.0003, 0.9951, 0.9546, 0.0090, 0.0051. These are descriptive indications of the refinement structure, not evidence of exact rank three or a test of all factorization constraints. The positive empirical likelihood difference rules out claiming exact equality of the fitted observed laws. A refinement likelihood-ratio comparison, with both models refined to the same numerical tolerance, gives a statistic of 18.26. In 199 parametric bootstrap replicates under M221, 17 statistics are at least this large, giving the plus-one tail probability $(17 + 1)/(199 + 1) = 0.09$ (Monte Carlo standard error approximately 0.02). No replicate violates the required likelihood nesting. At the conventional 5% threshold this comparison does not reject the refinement constraints; it does not establish equivalence or identify the true number of regional profiles. Residual dependence and other departures from the fitted model limit the calibration; SI S3.2 gives the protocol and Monte Carlo uncertainty.

The raw G5 Viterbi path has 9,123 runs, compared with 181 for M221. Much of this difference records switches between allele-fraction bands. Matching the Gaussian emissions and then grouping the paired G5 states gives 99.73% marker agreement with M221 and 194 runs. Thus a suitably aggregated ordinary HMM recovers nearly the same regional interpretation. The mixture makes the aggregation explicit in its transition model; it does not make that interpretation inaccessible to an ordinary HMM.

6.4 Prediction, replication and sensitivity

All held-out scores use fixed chromosome-1 parameters and sequential one-step predictive densities, with filtering restarted at each sequence boundary. Relative to G5, M221 gains 36.25 log-score units

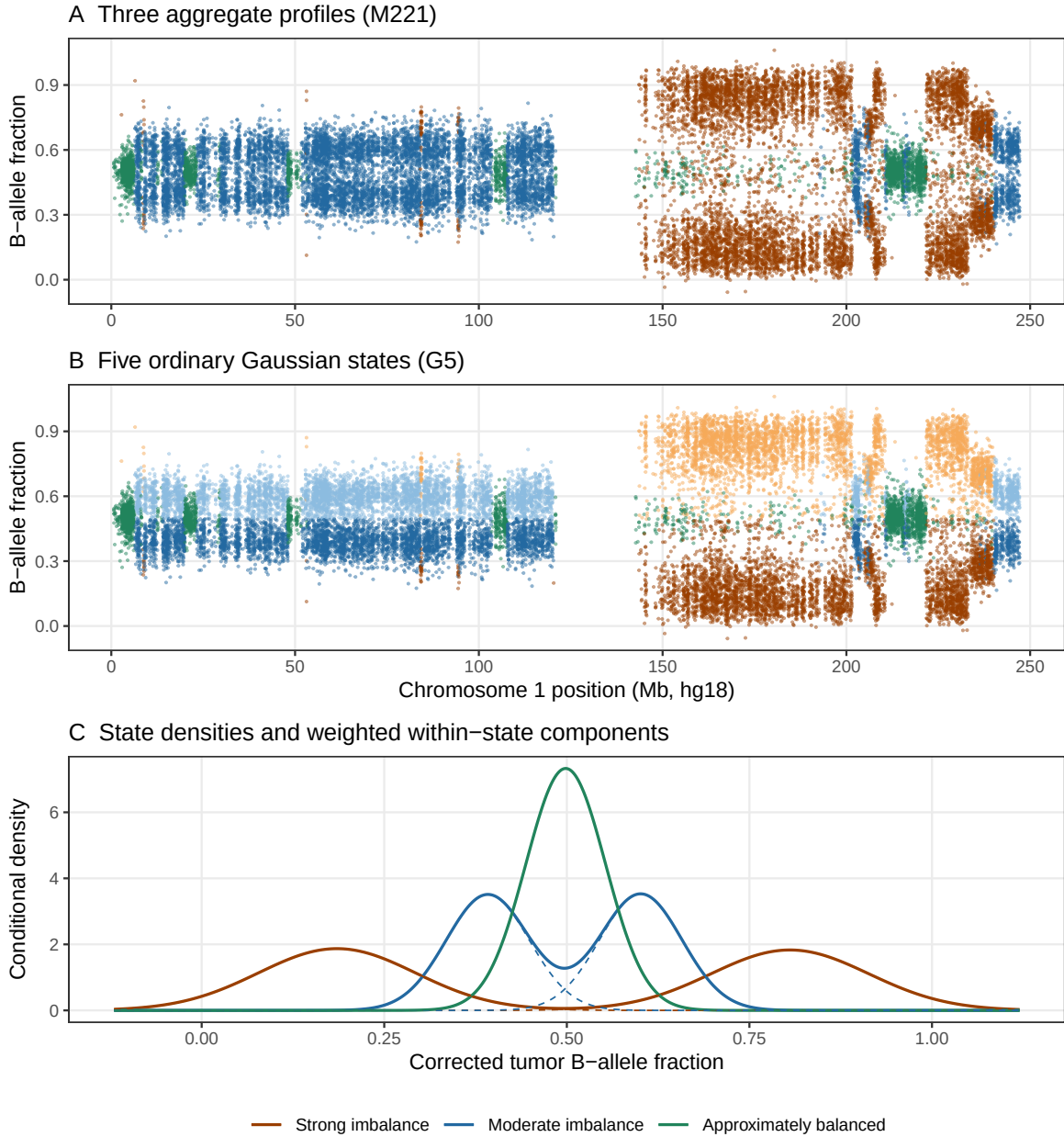


Figure 2: Chromosome 1 of HCC1143. Panels A and B colour the corrected fractions by the M221 and G5 Viterbi paths. The paired light and dark colours in B distinguish ordinary states that have the same aggregate colour in A. Panel C shows M221 state densities (solid) and weighted within-state component contributions (dashed). These are conditional densities, without multiplication by state occupation. The gap near the centromere resets the chain.

on chromosomes 12–17 and 5.16 on chromosomes 18–22, but loses 0.99 on the technical replicate. The corresponding differences per marker are 0.00069, 0.00020 and -0.00005 . These support similar predictive performance with fewer parameters, not a substantial predictive improvement or a formal equivalence test. No equivalence margin or sampling uncertainty for these score differences is estimated. The aggregated G5 path agrees with M221 on 99.07% and 99.06% of validation and confirmation markers. M221 assignments agree on 98.08% of markers shared between the two technical replicates. These are comparisons within one cell line, not independent patient-level validation.

F3 has the lowest BIC among the specified candidates, 17.4 below M221, although its training log likelihood is 21.15 lower. Its confirmation log score is 15.88 higher, with 14.97 of that difference arising on chromosome 20; its validation and technical-repeat scores are respectively 7.00 and 18.93 lower. Technical-repeat assignment agreement is 98.04%, compared with 98.08% for M221. Its estimated band separations reproduce the same three profiles. The symmetry constraint thus provides a parsimonious alternative, with predictive differences that vary across the evaluated sequences. The empirical purpose is to examine where an unrestricted HMM allocates complexity, rather than to propose a superior copy-number caller.

Predictive calibration is also qualified. M221’s central 95% predictive intervals cover 96.36% of validation and 95.48% of confirmation observations. The respective empirical PIT distances from uniformity are 0.055 and 0.012; Section S3 shows the histograms and residual diagnostics. Absolute residuals retain lag-one correlations of 0.057 and 0.142. We do not interpret near-nominal coverage as proof that all spatial dependence has been modeled.

Changing the standard-deviation floor to 0.005 or 0.02 leaves the M221 assignments unchanged, but both bounds are below the smallest fitted standard deviation (approximately 0.054), so these are inactive constraints. The two alternative normal-fraction filters give 98.58% and 98.76% agreement at shared markers, while changing the run count from 181 to 302 and 55, respectively. Retaining every second or fifth marker gives 97.56% and 96.35% agreement. However, the numbers of runs fall to 33 and 20 under thinning. The broad profiles are substantially more stable than the individual short segments. We therefore do not count all 181 runs as verified chromosomal alterations or interpret the transition probabilities as physical segment lengths.

A further check independently exchanges the B designation at each marker with probability one half, using a fixed random seed. Without refitting, M221 retains 99.87% of its assignments, compared with 56.63% for the raw G5 labels and 99.55% for aggregated G5 labels. F3 is exactly invariant by construction. M221 is only approximately invariant: its estimated weights and centres do not impose exact reflection symmetry.

6.5 Comparison with genomic segmentation and sequencing

We compare the chromosome-1 profiles with paired parent-specific circular binary segmentation (PSCBS; [Olshen et al., 2011](#)), which partitions a chromosome into regions with similar copy-number signals. It uses the same arrays’ total intensities and allele fractions. Its median segmented allelic-

imbalance measure, evaluated at markers assigned to the balanced, moderate and strong M221 profiles, is 0.080, 0.214 and 0.641. This supports the ordering of the profiles using an established segmentation procedure. It is not independent validation and uses an additional measurement that is absent from the HMM likelihood.

The published sequencing segments of [Chiang et al. \(2009\)](#) supply a separate assay of relative total copy ratio, on the same hg18 coordinate system. Between 210.542 and 221.671 Mb, 89.6% of the 1,152 retained markers receive the approximately balanced M221 label, while sequencing copy ratios in that interval range from 1.438 to 2.255 ([Figure 3](#)). An approximately balanced allele fraction therefore coexists with increased relative DNA dosage. The low- and high-fraction bands on either side instead form a sustained imbalance profile. This example makes the inferential distinction concrete: an allele-fraction component is not an additional regional regime, and a balanced regional profile does not establish a diploid copy number (two copies in total). The sequencing estimates corroborate this distinction; they do not provide error-free labels of allelic balance or establish segmentation accuracy.

7 Discussion

The theoretical and empirical results clarify why additional HMM states can be both useful and scientifically misleading. They are useful when they improve the approximation of a restrictive observation model. They are misleading when approximation components are labelled as new regimes without testing plausible emission alternatives.

The equal-component comparisons connect the empirical models to the exact refinement result. Constraining the refined transition matrix groups emission components into aggregate regimes; relaxing those constraints allows distinct component-level persistence. In the genomic analysis, nearly identical component distributions yield different raw state counts but very similar regional profiles after aggregation. The reduction from five ordinary states to three mixture states is supported by near-equality of paired transition rows, comparable held-out prediction and the allele-measurement mechanism. The model-conditional bootstrap comparison does not reject the refinement constraints at 5%, but this nonrejection is not an equivalence result. The symmetry-aware model has a lower BIC and mixed predictive differences across held-out sequences. These results illustrate the theoretical distinction without establishing a uniformly preferable genomic model. Decoded paths describe fitted partitions rather than an automatic state-merging rule.

Several limitations are important. Extra states can correct emissions, transitions, missing covariates, temporal resolution, or multiple mechanisms at once. Equality or near-equality of observed-law representations cannot identify a unique latent ontology. Domain knowledge, robustness to covariates and dependence structures, and scientific validation are therefore necessary ([Pohle et al., 2017](#); [McClintock et al., 2020](#)). The theorem does not prove finite-sample recovery or consistency of BIC selection. It is a one-way warning: under its assumptions, the aggregate-state simple-emission model is necessarily misspecified.

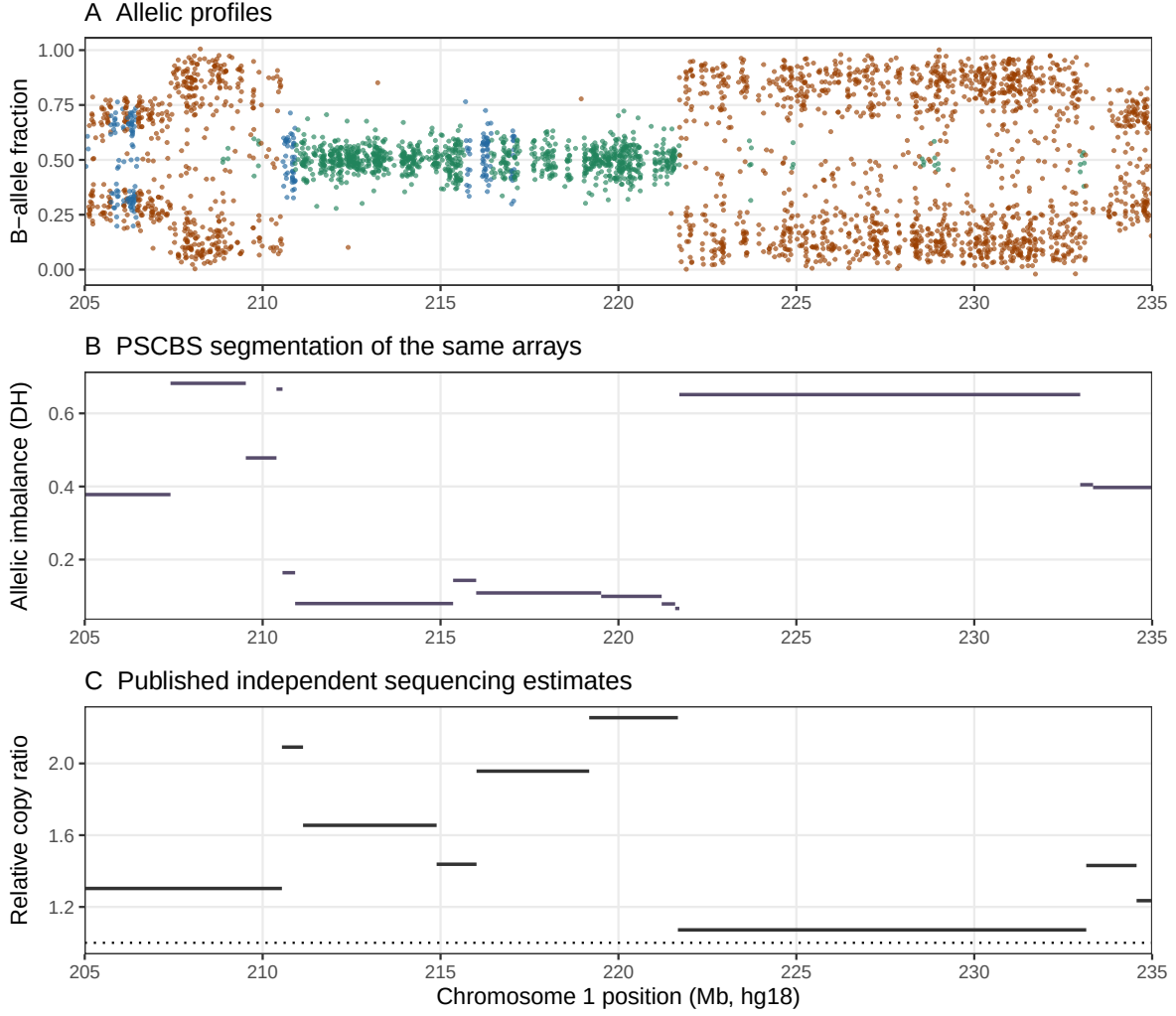


Figure 3: A chromosome-1 interval in the development data. Panel A shows M221 allele-fraction profiles, using the colours of Figure 2. Panel B shows the allelic-imbalance measure from PSCBS applied to the same arrays. Panel C shows the published independent sequencing copy-ratio segments of Chiang et al. (2009). Allelic balance and total dosage measure different properties. This development-data comparison is not a held-out classification test.

The exact result assumes finite mixtures and a positive aggregate transition matrix. Heavy-tailed or skewed emissions may require successively richer approximations rather than a finite exact refinement. Structural zeros in the transition matrix would require blockwise accessibility conditions instead of the uniform one-step bound used here.

The genomic analysis concerns one cell line and fixed candidate families. Chromosomes and technical repeats do not substitute for biological replication. Unequal marker spacing, residual dependence and sensitivity of short runs limit fine-scale segmentation. The independent sequencing assay measures total copy ratio, not the latent allelic profiles, so it cannot establish their true number or classification accuracy. The supplementary elk analyses additionally show that a

favorable Gaussian comparison can weaken under more appropriate emission distributions or temporal validation. Neither the theorem nor these examples identify a unique set of scientific regimes.

8 Conclusion

Emission misspecification can make additional hidden states valuable for observed-law approximation even when the scientific process has fewer aggregate regimes. Exact refinement and predictive mixture separation explain the mechanism. Comparing simple additional-state models with enriched-emission models at equal component count distinguishes additional transition freedom from additional emission flexibility. In the genomic data, this comparison recovers interpretable regional allelic profiles with nearly unchanged prediction and fewer transition parameters. Scientific interpretation requires evidence beyond the number of fitted states or density modes.

Acknowledgments

OpenAI ChatGPT and Codex were used in preparing and revising the text and computational materials, including code development, numerical checks and editorial assistance. Responsibility for the mathematical arguments, interpretation and final content remains with the author.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability statement

The SNP-array data are publicly available in Gene Expression Omnibus, accession GSE13372, at <http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE13372>. Sequencing segment estimates are in the supplementary materials of Chiang et al. (2009). The accompanying reproducibility package contains processed marker tables, fitted models, scripts, fixed seeds, source hashes and numerical tests. It documents regeneration from the original CEL files and annotation files, as well as reproduction from the supplied processed inputs. The supplementary elk analyses include their public-source provenance. The simulation code repository is <https://github.com/AurelienNicotiaULaval/corrective-hmm-states>; the accompanying package contains the analyses reported here.

Ethics statement

This study reanalyzes publicly available cell-line measurements and previously published animal data. No new participants were recruited, no new biological samples were collected and no animals were handled for this study.

Conflict of interest statement

The author declares no conflict of interest.

References

- Alexandrovich, G., H. Holzmann, and A. Leister (2016), Nonparametric identification and maximum likelihood estimation for hidden Markov models, *Biometrika*, 103(2), 423–434. doi: 10.1093/biomet/asw001.
- Allman, E. S., C. Matias, and J. A. Rhodes (2009), Identifiability of parameters in latent structure models with many observed variables, *The Annals of Statistics*, 37(6A), 3099–3132. doi: 10.1214/09-AOS689.
- Bengtsson, H., P. Wirapati, and T. P. Speed (2009), A single-array preprocessing method for estimating full-resolution raw copy numbers from all Affymetrix genotyping arrays including GenomeWideSNP 5 & 6, *Bioinformatics*, 25(17), 2149–2156. doi: 10.1093/bioinformatics/btp371.
- Bengtsson, H., P. Neuvial, and T. P. Speed (2010), TumorBoost: Normalization of allele-specific tumor copy numbers from a single pair of tumor-normal genotyping microarrays, *BMC Bioinformatics*, 11(1), 245. doi: 10.1186/1471-2105-11-245.
- Bickel, P. J., Y. Ritov, and T. Rydén (1998), Asymptotic normality of the maximum-likelihood estimator for general hidden Markov models, *The Annals of Statistics*, 26(4), 1614–1635. doi: 10.1214/aos/1024691255.
- Cappé, O., É. Moulines, and T. Rydén (2005), *Inference in Hidden Markov Models*, Springer. doi: 10.1007/0-387-28982-8.
- Chen, W., A. J. Robertson, D. Ganesamoorthy, and L. J. Coin (2017), sCNPhase: using haplotype resolved read depth to genotype somatic copy number alterations from low cellularity aneuploid tumors, *Nucleic Acids Research*, 45(5), e34. doi: 10.1093/nar/gkw1086.
- Chiang, D. Y., G. Getz, D. B. Jaffe, M. J. T. O’Kelly, X. Zhao, S. L. Carter, C. Russ, C. Nusbaum, M. Meyerson, and E. S. Lander (2009), High-resolution mapping of copy-number alterations with massively parallel sequencing, *Nature Methods*, 6(1), 99–103. doi: 10.1038/nmeth.1276.
- Dannemann, J. (2012), Semiparametric hidden Markov models, *Journal of Computational and Graphical Statistics*, 21(3), 677–692. doi: 10.1080/10618600.2012.681264.

- Dannemann, J., H. Holzmann, and A. Leister (2014), Semiparametric hidden Markov models: identifiability and estimation, *WIREs Computational Statistics*, 6(6), 418–425. doi: 10.1002/wics.1326.
- Douc, R. and É. Moulines (2012), Asymptotic properties of the maximum likelihood estimation in misspecified hidden Markov models, *The Annals of Statistics*, 40(5), 2697–2732. doi: 10.1214/12-AOS1047.
- Drton, M. and M. Plummer (2017), A Bayesian information criterion for singular models, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 79(2), 323–380. doi: 10.1111/rssb.12187.
- Gassiat, E. and C. Keribin (2000), The likelihood ratio test for the number of components in a mixture with Markov regime, *ESAIM: Probability and Statistics*, 4, 25–52. URL https://www.numdam.org/item/PS_2000__4__25_0/.
- Gassiat, E. and J. Rousseau (2014), About the posterior distribution in hidden Markov models with unknown number of states, *Bernoulli*, 20(4), 2039–2075. doi: 10.3150/13-BEJ550.
- Gassiat, E., A. Cleynen, and S. Robin (2016), Inference in finite state space nonparametric hidden Markov models and applications, *Statistics and Computing*, 26(1-2), 61–71. doi: 10.1007/s11222-014-9523-8.
- Holzmann, H. and F. Schwaiger (2015), Hidden Markov models with state-dependent mixtures: minimal representation, model testing and applications to clustering, *Statistics and Computing*, 25(6), 1185–1200. doi: 10.1007/s11222-014-9481-1.
- Keribin, C. (2000), Consistent estimation of the order of mixture models, *Sankhyā: The Indian Journal of Statistics, Series A*, 62(1), 49–66. URL <https://www.jstor.org/stable/25051289>.
- Langrock, R., T. Kneib, A. Sohn, and S. L. DeRuiter (2015), Nonparametric inference in hidden Markov models using P-splines, *Biometrics*, 71(2), 520–528. doi: 10.1111/biom.12282.
- Langrock, R., T. Adam, V. Leos-Barajas, S. Mews, D. L. Miller, and Y. P. Papastamatiou (2018), Spline-based nonparametric inference in general state-switching models, *Statistica Neerlandica*, 72(3), 179–200. doi: 10.1111/stan.12133.
- Lehéricy, L. (2019), Consistent order estimation for nonparametric hidden Markov models, *Bernoulli*, 25(1), 464–498. doi: 10.3150/17-BEJ993.
- Leroux, B. G. (1992), Maximum-likelihood estimation for hidden Markov models, *Stochastic Processes and their Applications*, 40(1), 127–143. doi: 10.1016/0304-4149(92)90141-C.
- Lin, Y. and M. Huang (2025), Penalized composite likelihood estimation for hidden Markov models with unknown number of states, *Statistics & Probability Letters*, 216, 110247. doi: 10.1016/j.spl.2024.110247.
- McClintock, B. T., R. Langrock, O. Gimenez, E. Cam, D. L. Borchers, R. Glennie, and T. A. Patterson (2020), Uncovering ecological state dynamics with hidden Markov models, *Ecology Letters*, 23(12), 1878–1903. doi: 10.1111/ele.13610.

- McLachlan, G. J. and D. Peel (2000), *Finite Mixture Models*, John Wiley & Sons. doi: 10.1002/0471721182.
- Olshen, A. B., H. Bengtsson, P. Neuvial, P. T. Spellman, R. A. Olshen, and V. E. Seshan (2011), Parent-specific copy number in paired tumor–normal studies using circular binary segmentation, *Bioinformatics*, 27(15), 2038–2046. doi: 10.1093/bioinformatics/btr329.
- Pohle, J., R. Langrock, F. M. van Beest, and N. M. Schmidt (2017), Selecting the number of states in hidden Markov models: pragmatic solutions illustrated using animal movement, *Journal of Agricultural, Biological and Environmental Statistics*, 22(3), 270–293. doi: 10.1007/s13253-017-0283-8.
- Rabiner, L. R. (1989), A tutorial on hidden Markov models and selected applications in speech recognition, *Proceedings of the IEEE*, 77(2), 257–286. doi: 10.1109/5.18626.
- Rousseau, J. and K. Mengersen (2011), Asymptotic behaviour of the posterior distribution in over-fitted mixture models, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(5), 689–710. doi: 10.1111/j.1467-9868.2011.00781.x.
- Teicher, H. (1963), Identifiability of finite mixtures, *The Annals of Mathematical Statistics*, 34(4), 1265–1269. doi: 10.1214/aoms/1177703862.
- Volant, S., C. Bérard, M.-L. Martin-Magniette, and S. Robin (2014), Hidden Markov models with mixtures as emission distributions, *Statistics and Computing*, 24(4), 493–504. doi: 10.1007/s11222-013-9383-7.
- Watanabe, S. (2013), A widely applicable Bayesian information criterion, *Journal of Machine Learning Research*, 14, 867–897. URL <https://www.jmlr.org/papers/v14/watanabe13a.html>.
- White, H. (1982), Maximum likelihood estimation of misspecified models, *Econometrica*, 50(1), 1–25. doi: 10.2307/1912526.
- Yakowitz, S. J. and J. D. Spragins (1968), On the identifiability of finite mixtures, *The Annals of Mathematical Statistics*, 39(1), 209–214. doi: 10.1214/aoms/1177698520.
- Zou, Y., Y. Lin, and X. Song (2024), Bayesian heterogeneous hidden Markov models with an unknown number of states, *Journal of Computational and Graphical Statistics*, 33(1), 15–24. doi: 10.1080/10618600.2023.2231055.
- Zucchini, W., I. L. MacDonald, and R. Langrock (2016), *Hidden Markov Models for Time Series: An Introduction Using R*, 2nd edition, Chapman and Hall/CRC. doi: 10.1201/b20790.

Figure legends

Figure 1. Observed histogram and fitted marginal densities for the reference simulation. The two-state Gaussian model smooths over the heterogeneous first regime. The three-state Gaussian and enriched two-state models recover its two components.

Figure 2. Chromosome 1 of HCC1143. Panels A and B show the M221 and G5 Viterbi labels. Paired light and dark colours in B share an aggregate colour in A. Panel C separates conditional state densities and weighted within-state component contributions. The chain resets across the gap near the centromere.

Figure 3. Development-data comparison of M221 allele-fraction profiles, PSCBS allelic imbalance from the same arrays, and independent sequencing copy-ratio estimates. An approximately balanced fraction does not identify unchanged total DNA dosage.